

SMRTTECH 4AI3 - Artificial Intelligence and Machine Learning

Lab 2 - Linear Regression

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Due: Midnight 11:59PM, day before your next lab session

Please read and follow these instructions:

- Submit your lab on Avenue to Learn (A2L) before the due date (midnight of the day before your next lab session). The submission will be closed after the deadline.
- Do your lab exercises on your own, i.e., do not copy. If two lab reports (or some parts of them) are found to be copies of each other, then BOTH shall receive 0 mark.
- You may use Google Colab or any other Python programming platform of your own choice to run your programs.
- Please read the questions carefully and use AI / machine learning libraries only when specially required as part of the question. Also add comments in your code where ever necessary.
- Submit your working python source files (or Jupyter notebooks) in zipped or source code format.

Exercises (20 Points)

In this lab, you will perform linear regression using the **honeyproduction** dataset. You will implement linear regression from scratch and compare it with the linear regression model provided by **scikit-learn (sklearn)**. Finally, you will compare the results of your custom implementation with that of sklearn.

1. **Import Required Libraries:** Begin by importing the necessary libraries such as numpy, pandas, matplotlib, and sklearn for the analysis. **(1 Point)**
2. **Upload and Preview the Dataset:** Upload the honeyproduction dataset. Use head() to display the first 5 rows of the dataset for an initial overview. **(1 Point)**
3. **Inspect the Dataset:** Check the dataset's structure using the info() function to understand its columns, data types, and any missing values. **(1 Point)**
4. **Calculate the Average Price Per Year:** Group the data by year and calculate the average priceperlb (price per pound of honey). Create a new table from this aggregated data. **(1 Point)**
5. **Prepare Feature and Target Variables:** Select Year as the feature (input variable X). Select priceperlb as the target (output variable Y). **(1 Point)**

6. **Visualize the Data:** Plot a scatter plot with Year on the x-axis and priceperlb on the y-axis to visualize the relationship between the feature and target. **(1 Point)**
7. **Build Linear Regression Models:** Build two linear regression models **(2 Points)**
Model 1: Implement linear regression from scratch.
Model 2: Use scikit-learn's LinearRegression class to build the model.
8. **Train the Models:** Train both Model 1 and Model 2 on the training data (Year and priceperlb). **(2 Points)**
9. **Obtain the Trained Weights:** Extract the learned weights (slope and intercept) from both models after training. **(1 Point)**
10. **Make Predictions:** Use both models to make predictions on the training data (i.e., predict the price of honey based on Year). **(1 Point)**
11. **Visualize Predicted and True Values:** Plot both the predicted values and the true values on the same graph for both models. Use different colors or markers to distinguish between the models and the true data points. **(2 Points)**
12. **Examine the Hypothesis Lines:** Write down the hypothesis/line equations for both models. **(1 Point)**
13. **Make Future Predictions:** Use both trained models to predict the price of honey in the year 2024. **(1 Point)**
14. **Compare the Results:** Compare the performance and predictions of both models. **(4 Points)**
Discuss:
How similar or different the learned weights (slope and intercept) are between the two models.
How close the predictions are for both models, including the prediction for 2024.
Any differences in accuracy or other observations between your custom implementation and sklearn's model.